AP CALCULUS AB

Unit 1 • Lesson 1.2 practice • Answer key

Lesson 1.2 • Defining Limits and Using Limit Notation

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What does $\lim_{x \to 2} f(x) = 5$ mean?

(A) $f(2) = 5$.

(B) $f(x)$ gets as close to 5 as we want when x is close enough to 2, $x \neq 2$.

(C) $f(x)$ never equals 5.

(D) $f(5) = 2$.

Solution

Answer: (B). A limit describes outputs near $x = 2$, not the value at $x = 2$.

Question 2

NO CALCULATOR

$\lim_{x \to 3^-} f(x) = 4$, $\lim_{x \to 3^+} f(x) = 4$ and $f(3) = 1$. What is $\lim_{x \to 3} f(x)$?

(A) 1

(B) 4

(C) 5

(D) It does not exist.

Solution

Answer: (B). Both one-sided limits equal 4. The value $f(3)$ does not matter.

Question 3

NO CALCULATOR

$\lim_{x \to 0^-} g(x) = 2$ and $\lim_{x \to 0^+} g(x) = 5$. Which statement is true?

(A) $\lim_{x \to 0} g(x) = 2$

(B) $\lim_{x \to 0} g(x) = 3.5$

(C) $\lim_{x \to 0} g(x)$ does not exist.

(D) $g(0) = 5$

Solution

Answer: (C). The one-sided limits are different.

Question 4

NO CALCULATOR

Let $f(x) = x^2$ for $x \neq 1$ and $f(1) = 7$. What is $\lim_{x \to 1} f(x)$?

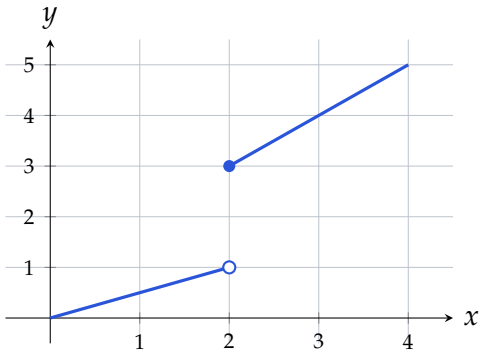
(A) 7
(B) 8
(C) It does not exist.
(D) 1

Solution
Answer: (D). Near $x = 1$ (but not at it), $f(x) = x^2$, which approaches 1.

Question 5 (graph)

NO CALCULATOR

The graph of f is shown. Which statement is true?



(A) $\lim_{x\to 2} f(x) = 3$
(B) $\lim_{x\to 2^-} f(x) = 1$
(C) $\lim_{x\to 2^-} f(x) = 3$
(D) $\lim_{x\to 2^+} f(x) = 1$

Solution
Answer: (B). From the left, the graph approaches the open circle at height 1. From the right it approaches 3, so $\lim_{x\to 2} f(x)$ does not exist.

Question 6 (Free response)

NO CALCULATOR

Let $f(x) = \begin{cases} x+2, & x < 1 \\ 5, & x = 1 \\ x^2+2, & x > 1. \end{cases}$

(a) Find $\lim\limits_{x\to 1^-} f(x)$ and $\lim\limits_{x\to 1^+} f(x)$.
(b) Does $\lim\limits_{x\to 1} f(x)$ exist? If so, give its value.
(c) Explain why the value $f(1) = 5$ does not change your answer to (b).

Solution
(a) $\lim_{x\to 1^-} f(x) = 1 + 2 = 3$ and $\lim_{x\to 1^+} f(x) = 1^2 + 2 = 3$.
(b) Yes. Both one-sided limits equal 3, so $\lim_{x\to 1} f(x) = 3$.
(c) A limit only uses values of $f(x)$ for x near 1 with $x \neq 1$; the value at $x = 1$ itself is not used.